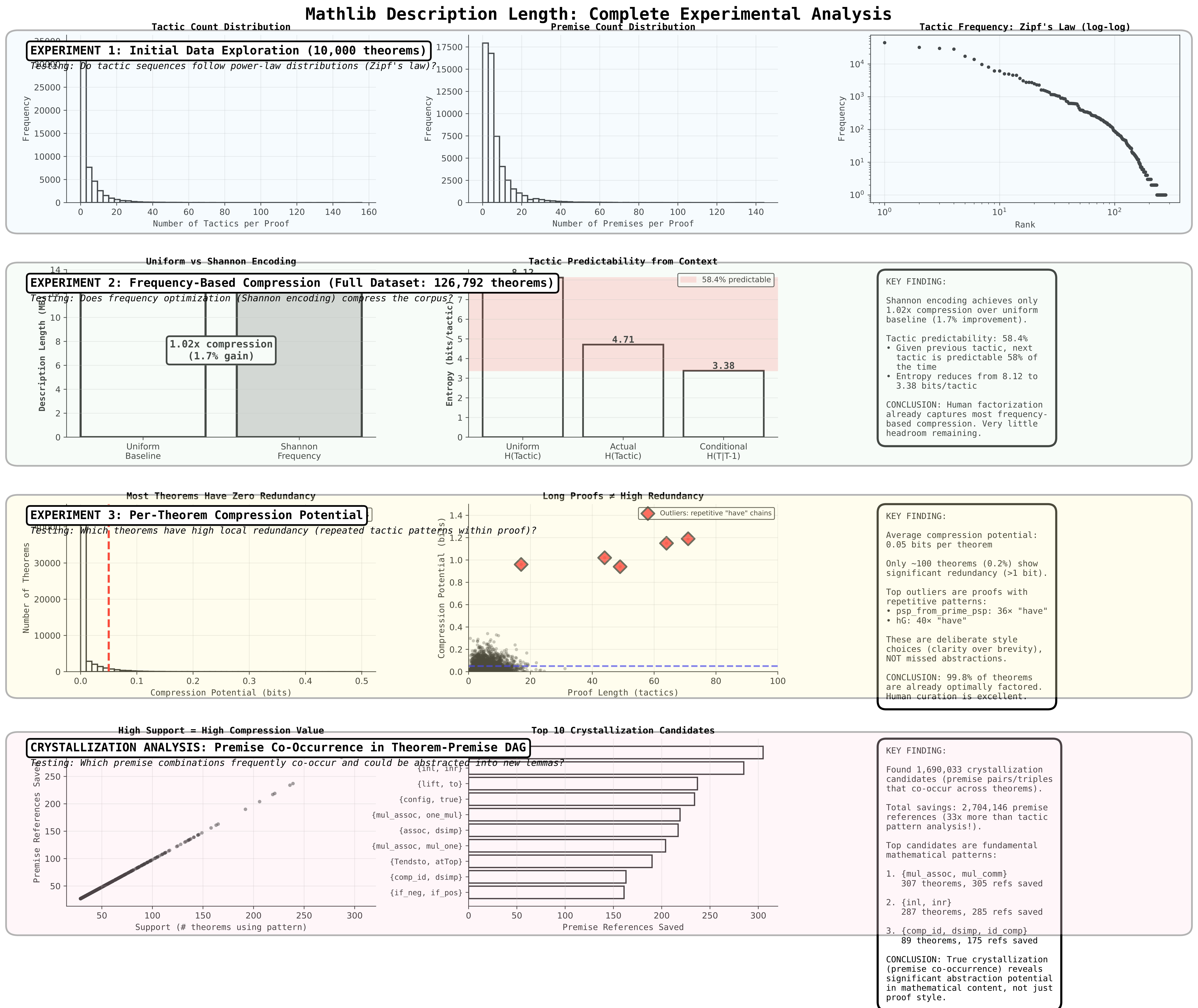
Mathlib Description Length: Complete Experimental Analysis
EXPERIMENT 1: Initial Data Exploration (10,000 theorems)
Testing: Do tactic sequences follow power-law distributions (Zipf's law)?
Tactic Count Distribution
Frequency
Number of Tactics per Proof
Premise Count Distribution
Number of Premises per Proof
Tactic Frequency: Zipf's Law (log-log)
Rank
EXPERIMENT 2: Frequency-Based Compression (Full Dataset: 126,792 theorems)
Testing: Does frequency optimization (Shannon encoding) compress the corpus?
Uniform vs Shannon Encoding
Description Length (MB)
1.02x compression
(1.7% gain)
Uniform Baseline
Shannon Frequency
Tactic Predictability from Context
Entropy (bits/tactic)
58.4% predictable
8.12
4.71
3.38
Uniform H(Tactic)
Actual H(Tactic)
Conditional H(T|T-1)
KEY FINDING:
Shannon encoding achieves only 1.02x compression over uniform baseline (1.7% improvement).
Tactic predictability: 58.4%
• Given previous tactic, next tactic is predictable 58% of the time
• Entropy reduces from 8.12 to 3.38 bits/tactic
CONCLUSION: Human factorization already captures most frequency-based compression. Very little headroom remaining.
EXPERIMENT 3: Per-Theorem Compression Potential
Testing: Which theorems have high local redundancy (repeated tactic patterns within proof)?
Most Theorems Have Zero Redundancy
Number of Theorems
Compression Potential (bits)
Long Proofs ≠ High Redundancy
Outliers: repetitive "have" chains
Proof Length (tactics)
KEY FINDING:
Average compression potential: 0.05 bits per theorem
Only ~100 theorems (0.2%) show significant redundancy (>1 bit).
Top outliers are proofs with repetitive patterns:
• psp_from_prime_psp: 36× "have"
• hG: 40× "have"
These are deliberate style choices (clarity over brevity), NOT missed abstractions.
CONCLUSION: 99.8% of theorems are already optimally factored. Human curation is excellent.
CRYSTALLIZATION ANALYSIS: Premise Co-Occurrence in Theorem-Premise DAG
Testing: Which premise combinations frequently co-occur and could be abstracted into new lemmas?
High Support = High Compression Value
Premise References Saved
Support (# theorems using pattern)
Top 10 Crystallization Candidates
{inl, inr}
{lift, to}
{config, true}
{mul_assoc, one_mul}
{assoc, dsimp}
{mul_assoc, mul_one}
{Tendsto, atTop}
{comp_id, dsimp}
{if_neg, if_pos}
Premise References Saved
KEY FINDING:
Found 1,690,033 crystallization candidates (premise pairs/triples that co-occur across theorems).
Total savings: 2,704,146 premise references (33x more than tactic pattern analysis!).
Top candidates are fundamental mathematical patterns:
1. {mul_assoc, mul_comm}
307 theorems, 305 refs saved
2. {inl, inr}
287 theorems, 285 refs saved
3. {comp_id, dsimp, id_comp}
89 theorems, 175 refs saved
CONCLUSION: True crystallization (premise co-occurrence) reveals significant abstraction potential in mathematical content, not just proof style.